

Module Theory

Exercise 1.

- (a) The element 2 is invertible in both $\mathbb{Z}[\frac{1}{2}]$ and in $\mathbb{Z}/3\mathbb{Z}$. Find a homomorphism $\phi: \mathbb{Z}[\frac{1}{2}] \rightarrow \mathbb{Z}/3\mathbb{Z}$ which sends $2^{-1} \mapsto 2^{-1}$.
- (b) Find an element in $\ker \phi$ which is not in $\mathbb{Z}$.

Solution 1.

- (a) In $\mathbb{Z}/3\mathbb{Z}$, $2^{-1} = 2$. So, we take the map that sends $\frac{n}{2^k}$ to $n \cdot 2^k \pmod{3}$.
- (b) The element $\frac{3}{2}$ is in $\ker \phi$ but not $\mathbb{Z}$.

Exercise 2. Recall that if R is a ring and $S \subset R$ is a multiplicative subset, then in the localization $S^{-1}R$ two fractions $\frac{r_1}{s_1}, \frac{r_2}{s_2}$ are equal to one another if there is some $s_3 \in S$ so that $s_3(r_1s_2 - r_2s_1) = 0$. For the rest of this problem, let $R = \mathbb{Z}/12\mathbb{Z}$, and consider the localizations R_f for various $f \in R$.

- (a) For a given f , which fractions $\frac{n}{f^k}$ are equal to $\frac{0}{1}$ in the localization?
- (b) Prove that for a given f , in R_f we have $\frac{1}{f} = \frac{y}{1}$ for some $y \in R$. Conclude that every $\frac{n}{f^k}$ is equal to a fraction with 1 in the denominator. (Hint: rephrase this in terms of something being equal to 0.)
- (c) Which rings can arise as localizations R_f ? For each possible ring, which f s achieve that ring?
- (d) Draw the lattice of ideals in R , and the lattice of ideals in each possible R_f . What do you notice comparing them?

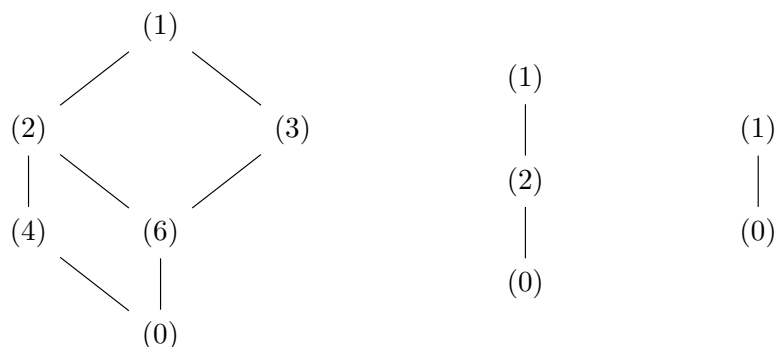
Solution 2.

- (a) By the definition of equality in the localization, $\frac{n}{f^k}$ is equal to 0 exactly when there is some f^i so that $f^i n = 0$ in $\mathbb{Z}/12\mathbb{Z}$. This can happen four ways: (1) $f \in \{0, 6\}$, (2) $n = 0$, (3) $2 \mid f$ and $3 \mid n$, or (4) $3 \mid f$ and $4 \mid n$. (Note, if f is coprime to 12, then only (2) can happen.)
- (b) Saying that $\frac{1}{f} = \frac{y}{1}$ is equivalent to saying that $\frac{yf-1}{f} = \frac{0}{1}$, and we already know which fractions are 0. So, correspondingly, we have four cases: (1) if $f \in \{0, 6\}$ then every fraction is equal to 0, so we can choose $y = 0$; (2) if f is coprime to 12, then f is invertible mod 12, so we can choose $y = f^{-1}$; (3) if $2 \mid f$ and $f \neq 6$, then either $3 \mid f - 1$ or $3 \mid 2f - 1$, so we can choose either $y = 1$ or $y = 2$ correspondingly; (4) similarly, if $3 \mid f$ and $f \neq 6$, then either $4 \mid f - 1$ or $4 \mid 3f - 1$, so we can choose either $y = 1$ or $y = 3$ correspondingly.

(c) From our previous computations, we get the table below.

f	R_f
0, 6	0
1, 5, 7, 11	$\mathbb{Z}/12\mathbb{Z}$
2, 4, 8, 10	$\mathbb{Z}/3\mathbb{Z}$
3, 9	$\mathbb{Z}/4\mathbb{Z}$

(d) Below are the lattices of ideals for $\mathbb{Z}/12\mathbb{Z}$, $\mathbb{Z}/4\mathbb{Z}$, and $\mathbb{Z}/3\mathbb{Z}$.



The prime ideals (2) and (3) are at the top just under (1). Notice how when we invert an element divisible by p , all the ideals generated by powers of p get smushed up to become the unit ideal. Also, notice that this process can smush other ideals together, e.g. when we invert 2, the ideals $(6) = (2)(3)$ and $(0) = (4)(3)$ become equal to the ideal (3), since the factors of (2) and (4) become factors of (1).

Exercise 3. (January 2020 Problem 3) Let R be a commutative ring, I an ideal in R , and S a multiplicative subset of R . There are two standard correspondences involving prime ideals:

- (i) Prime ideals in R/I are in bijection with prime ideals in R which contain I .
 - (ii) Prime ideals in $S^{-1}R$ are in bijection with prime ideals in R which do not meet S .
- (a) Prove (i) and (ii). (The original problem said “or”, but in the SEP you have no choice.)
- (b) Give an example of a ring with finitely many elements that has exactly three prime ideals.
- (c) Give an example of a subring of $\mathbb{Q}$ that has exactly three prime ideals.

Solution 3.

- (a) (i) Let P be a prime ideal which contains I . Then P/I is a proper ideal of R/I . Let $x, y \in R$ such that $[x][y] \in P/I$, where $[x]$ denotes the class of x in R/I . Then $xy = p + i$ for some $p \in P$ and $i \in I$. Since $P \supset I$, we in fact have $xy \in P$, which means $x \in P$ or $y \in P$. Thus, $[x] \in P/I$ or $[y] \in P/I$.
- On the other hand, let $Q \subset R/I$ be a prime ideal, and consider $P = \pi^{-1}(Q)$, where $\pi: R \rightarrow R/I$ is the quotient map. Then P is an ideal of R . Let $xy \in P$, then $[x][y] \in Q$, and so either $[x] \in Q$ or $[y] \in Q$. Thus, either $x \in P$ or $y \in P$.

- (ii) Let P be a prime ideal which does not meet S : $P \cap S = \emptyset$. Then $S^{-1}P$ is a proper ideal of $S^{-1}R$. Suppose $\frac{x}{s_1} \cdot \frac{y}{s_2} \in S^{-1}P$, then there exist $p \in P$ and $s_3 \in S$ such that $xy s_3 = p s_1 s_2$. Notice here that the RHS is in P , so $xy s_3 \in P$. But $s_3 \notin P$, so $xy \in P$, and thus either $x \in P$ or $y \in P$. Therefore, either $\frac{x}{s_1} \in S^{-1}P$ or $\frac{y}{s_2} \in S^{-1}P$.

On the other hand, let $Q \subset S^{-1}R$ be a prime ideal, and let $P = \ell^{-1}(Q)$, where $\ell: R \rightarrow S^{-1}R$ is the localization map. From here, the argument is identical to the above.

- (b) $\mathbb{Z}/30\mathbb{Z}$ works. You need some integer n so that (n) is contained in exactly three prime ideals, which is the same as saying that n has exactly three distinct prime factors.
- (c) Let $S = \{n \in \mathbb{Z} : 2 \nmid n, 3 \nmid n\}$. Then $S^{-1}\mathbb{Z} \subset \mathbb{Q}$ has only the prime ideals (0) , (2) , and (3) . There's a subtlety here which is that we should only design our S to miss *two* prime ideals, because our S will always miss the 0 ideal, which is prime since $\mathbb{Z}$ is an integral domain.

Exercise 4.

- (a) Prove that *localization is exact*. That is, if $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ is an exact sequence of R -modules, then $0 \rightarrow S^{-1}A \xrightarrow{S^{-1}f} S^{-1}B \xrightarrow{S^{-1}g} S^{-1}C \rightarrow 0$ is an exact sequence of $S^{-1}R$ -modules.
- (b) If B is an R -module, and A is a submodule, prove that $S^{-1}(B/A) \cong S^{-1}B/S^{-1}A$. That is, you can localize or quotient in any order.
- (c) Compute the isomorphism types of the following:

$$\left(\frac{\mathbb{C}[x, y]}{(xy)} \right)_x \quad \text{and} \quad \left(\frac{\mathbb{C}[x, y]}{(xy)} \right)_{x+y}.$$

(Hint for the second one: show that $(x)_{x+y} + (y)_{x+y} = (1)$ and use the Chinese Remainder Theorem.)

Solution 4.

- (a) We will proceed in three parts:

Exactness at $S^{-1}C$: We want to show that $S^{-1}g$ is surjective. Suppose $c/s \in S^{-1}C$, then we can find $b \in B$ so that $g(b) = c$. Then $S^{-1}g(b/s) = g(b)/s = c/s$.

Exactness at $S^{-1}B$: We want to show that $\ker S^{-1}g = \text{im } S^{-1}f$. Suppose b/s is in $\ker S^{-1}g$. Let $c = g(b)$, then $c/s = 0$ means there is some $s' \in S$ so that $s'c = 0$. Thus, $s'b \in \ker g = \text{im } f$, so there is some $a \in A$ so that $f(a) = s'b$. Then $S^{-1}f(a/(ss')) = (s'b)/(ss') = b/s$, so $b/s \in \text{im } S^{-1}f$.

Exactness at $S^{-1}A$: We want to show that $S^{-1}f$ is injective. Suppose $a/s \in \ker S^{-1}f$. Let $b = f(a)$, then similar to the above there is some $s' \in S$ so that $s'b = 0$. Thus, $s'a = 0$ since f is injective. But then $a/s = 0$ in $S^{-1}A$ to begin with.

- (b) Note that $S^{-1}(B/A) = S^{-1}C$, and by part (a) $S^{-1}C = (S^{-1}B)/(S^{-1}A)$, as desired.

(c) For the first one, we have

$$\begin{aligned} \left(\frac{\mathbb{C}[x, y]}{(xy)} \right)_x &\cong \frac{\mathbb{C}[x, y]_x}{(xy)_x} \\ &\cong \frac{\mathbb{C}[x, y]_x}{(y)_x} \\ &\cong \left(\frac{\mathbb{C}[x, y]}{(y)} \right)_x \\ &\cong \mathbb{C}[x]_x \cong \mathbb{C}[x, x^{-1}]. \end{aligned}$$

For the second one, first note that $(x)_{x+y} + (y)_{x+y} = (x, y)_{x+y} = (1)$, since $x + y \in (x, y)$. Thus,

$$\begin{aligned} \left(\frac{\mathbb{C}[x, y]}{(xy)} \right)_{x+y} &\cong \frac{\mathbb{C}[x, y]_{x+y}}{(xy)_{x+y}} \\ &\cong \frac{\mathbb{C}[x, y]_{x+y}}{(y)_{x+y}} \times \frac{\mathbb{C}[x, y]_{x+y}}{(x)_{x+y}} && \text{(by CRT)} \\ &\cong \left(\frac{\mathbb{C}[x, y]}{(y)} \right)_{x+y} \times \left(\frac{\mathbb{C}[x, y]}{(x)} \right)_{x+y} \\ &\cong \mathbb{C}[x]_x \times \mathbb{C}[y]_y \cong \mathbb{C}[x, x^{-1}] \times \mathbb{C}[y, y^{-1}]. \end{aligned}$$

Exercise 5.

- (a) Prove that a $\mathbb{C}[x]$ -module is the same thing as a $\mathbb{C}$ -vector space V and a linear transformation $X: V \rightarrow V$.
- (b) Suppose $V = \mathbb{C}^n$, and $X: V \rightarrow V$ is a linear transformation. Then (V, X) is a finitely-generated $\mathbb{C}[x]$ -module, so it is isomorphic to $\bigoplus_i \mathbb{C}[x]/(f_i(x))$ for some polynomials $f_i(x)$. In linear algebra terms, what is the relation between the polynomials f_i and the linear transformation X ?

Solution 5.

- (a) For any $\mathbb{C}[x]$ -module M , we can multiply by scalars in $\mathbb{C}$, so M is also a $\mathbb{C}$ -vector space. Multiplication by x commutes with multiplication by scalars, so x induces a linear transformation $M \rightarrow M$. On the other hand, if V is a $\mathbb{C}$ -vector space, and $X: V \rightarrow V$ is a linear transformation, then we can make V into a $\mathbb{C}[x]$ -module via $f(x) \cdot v = f(X)v$.
- (b) If the f_i are all powers of irreducible polynomials, then over $\mathbb{C}$ the f_i all look like $(x - \lambda_i)^{e_i}$. The λ_i are of course the eigenvalues of the linear transformation X , the powers e_i are the sizes of the Jordan blocks in the JNF of X , the product of the f_i is the characteristic polynomial of X , and the lcm of the f_i is the minimal polynomial of X .

Exercise 6. (January 2022 Problem 3) Let R be a commutative ring.

- (a) Prove that a module M over R is projective if and only if it is a direct summand of a free module.
- (b) Let M be a projective module over R and $S \subset R$ a multiplicative set. Prove that $S^{-1}M$ is a projective $R[S^{-1}]$ -module.

Solution 6.

- (a) For the forward direction, assume M is projective, and let $\pi: F \rightarrow M$ be a surjective map where F is a free module. Such a map exists since every module is a homomorphic image of a free module. Then by definition of M being projective, there exists a map $\varphi: M \rightarrow F$ such that $\pi \circ \varphi = id_M$. Therefore, the short exact sequence $0 \rightarrow \ker \pi \rightarrow F \xrightarrow{\pi} M \rightarrow 0$ splits, and thus $F \cong M \oplus \ker \pi$.

For the reverse direction, assume $F \cong M \oplus N$ for some free module F . Then there exist natural maps $\alpha: M \rightarrow F$ and $\beta: F \rightarrow M$ given by inclusion and projection such that $\beta \circ \alpha = id_M$. Suppose we have a surjective map $\pi: L \rightarrow K$ and a map $\psi: M \rightarrow K$. We need to show that

there exists a map $\varphi: M \rightarrow L$ such that the following commutes:

$$\begin{array}{ccc} & M & \\ \swarrow \varphi & \downarrow \psi & \\ L & \xrightarrow{\pi} & K \end{array}$$

Since F is free, it

is projective (this is not hard to prove using that F has a basis and so any map out of F is defined by what it does to the basis elements). Therefore, considering the map $\psi \circ \beta: F \rightarrow M$, we know that there exists a map $\gamma: F \rightarrow L$ such that $\pi \circ \gamma = \psi \circ \beta$. Then defining $\varphi = \gamma \circ \alpha: M \rightarrow L$, we see that $\pi \circ \varphi = \pi \circ \gamma \circ \alpha = \psi \circ \beta \circ \alpha = \psi$. Thus, M is projective.

- (b) Since M is projective, by part (a), we have that $M \oplus N \cong R^d$ for some R -module N and some $d \geq 1$. Therefore, we have a split SES $0 \rightarrow N \rightarrow R^d \rightarrow M \rightarrow 0$. Since localization is exact, we get that $0 \rightarrow S^{-1}N \rightarrow S^{-1}(R^d) \rightarrow S^{-1}M \rightarrow 0$ is also a split SES. Thus, $S^{-1}M \oplus S^{-1}N \cong S^{-1}(R^d) \cong R[S^{-1}]^d$, so $S^{-1}M$ is a projective $R[S^{-1}]$ -module by part (a).

Exercise 7. (January 2021 Problem 3) Two questions about projective modules.

- (a) Show that a projective module over an integral domain is torsion free.
- (b) Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of finitely generated modules over a commutative, Noetherian ring. Prove the following OR give a counterexample:
- (i) If M_1 and M_2 are projective, then so is M_3 .
 - (ii) If M_2 and M_3 are projective, then so is M_1 .

Solution 7.

- (a) Let M be a projective R -module, where R is an integral domain. Then M is a direct summand of a free R -module, i.e. $M \oplus N = R^d$ for some R -module N and $d \in \mathbb{N}$. Let $m \in M$ be a nonzero element, and suppose that $am = 0$ for some $a \in R$. We will show that $a = 0$. We can write $m = (r_1, \dots, r_d)$ with $r_i \in R$. Then $am = (ar_1, \dots, ar_d) = 0$ implies that $ar_i = 0$ in R for all i . Since R is a domain, this means that either $a = 0$ or $r_i = 0$ for all i . Since $m \neq 0$, we must have that $a = 0$. Thus, M is torsion free.
- (b) (i) This is not true. For example, consider the SES of $\mathbb{Z}$ -modules $0 \rightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$, where the left map is multiplication by 2 and the right map is the natural quotient map. (Or similarly, $0 \rightarrow k[x] \xrightarrow{\cdot x} k[x] \rightarrow k \rightarrow 0$.) Then $\mathbb{Z}$ is projective, but $\mathbb{Z}/2\mathbb{Z}$ is not a projective

$\mathbb{Z}$ -module. One way to see this directly is that if $\pi : \mathbb{Z}/4\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is the map given by modding out by 2 and $\mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is the identity map, then there is no map that makes the following diagram commute:

$$\begin{array}{ccc} & \mathbb{Z}/2\mathbb{Z} & \\ \swarrow \text{dashed} & \downarrow & \\ \mathbb{Z}/4\mathbb{Z} & \longrightarrow & \mathbb{Z}/2\mathbb{Z} \end{array}$$

This because the only possible maps are the zero map and multiplication by 2, and neither of these gives the identity map when composed with π . Another way to see that $\mathbb{Z}/2\mathbb{Z}$ is not a projective $\mathbb{Z}$ -module is to use that every projective module is flat and show that it is not flat.

- (ii) This is true. If M_3 is projective, then the SES is split exact, so $M_2 \cong M_1 \oplus M_3$. Recall that being projective is equivalent to being a direct summand of a free module. If M_2 is projective, then M_1 is as well since it is a direct summand of M_2 which is a direct summand of a free module.

Exercise 8. (January 2025 Problem 5, parts (a), (c), (d)) Let A be a commutative Noetherian ring, and M an A -module. We say that M has finite length if there exists a positive integer n such that every chain

$$M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_k$$

satisfies $k \leq n$. The smallest such n is then called the length of M .

- (a) Prove that if M has finite length, then so does every localization $M_{\mathfrak{p}}$ at a prime ideal $\mathfrak{p}$ of A .
- (c) Prove that the following converse of (a) holds if A has finitely many maximal ideals: under this assumption, M has finite length if $M_{\mathfrak{m}}$ has finite length for every maximal ideal $\mathfrak{m}$ of A . (You can use without proof the fact that an A -module M is zero if and only if $M_{\mathfrak{m}}$ is zero for every maximal ideal $\mathfrak{m}$ of A .)
- (d) Prove that the assumption in part (c) on the finiteness of the number of maximal ideals is necessary by providing an example of a ring A and an A -module M which is not of finite length, even though $M_{\mathfrak{m}}$ is of finite length for every maximal ideal $\mathfrak{m}$ of A . (Hint: you can find such an example where A is the ring of integers.)

Solution 8.

- (a) Given a prime $\mathfrak{p} \subset A$, we will show that $\text{len}(M_{\mathfrak{p}}) \leq \text{len}(M)$. Let $\ell : M \rightarrow M_{\mathfrak{p}}$ be the localization map. Given any chain contained in $M_{\mathfrak{p}}$, $\widetilde{M}_0 \subsetneq \cdots \subsetneq \widetilde{M}_k$, we want to show that M contains a chain at least as long. Define $M_i = \ell^{-1}(\widetilde{M}_i) \subset M$. Clearly $M_i \subseteq M_{i+1}$ for all i , so if we show that the inclusions are strict we will be done.

Since $\widetilde{M}_i \subsetneq \widetilde{M}_{i+1}$, we can find some $m/s \in \widetilde{M}_{i+1} \setminus \widetilde{M}_i$ where $s \notin \mathfrak{p}$. If we look in the quotient $\widetilde{M}_{i+1}/\widetilde{M}_i$, the annihilator of the class $[m/s]$ is an ideal of $A_{\mathfrak{p}}$. Since $A_{\mathfrak{p}}$ is a local ring, and since the class $[m/s]$ was chosen not to be the zero class, the annihilator must be contained in the maximal ideal $\mathfrak{p}$. Since $s \notin \mathfrak{p}$, s does not annihilate $[m/s]$, so $s[m/s] \neq [0]$ in the quotient, i.e. $s(m/s) = m/1 \notin \widetilde{M}_i$. Hence, $m/1 \in \widetilde{M}_{i+1} \setminus \widetilde{M}_i$, and so $m \in M_{i+1} \setminus M_i$.

- (b) For each maximal ideal $\mathfrak{m} \subset A$, let $\text{len}(M_{\mathfrak{m}}) = n_{\mathfrak{m}}$. We claim $\text{len}(M) \leq \sum_{\mathfrak{m}} n_{\mathfrak{m}}$. Let $M_0 \subsetneq \cdots \subsetneq M_k$ be a chain. Localizing at a maximal ideal $\mathfrak{m}$, we find there are at most $n_{\mathfrak{m}}$ locations where $(M_i)_{\mathfrak{m}} \subsetneq (M_{i+1})_{\mathfrak{m}}$, and the other inclusions must be equality. If k were greater than $\sum_{\mathfrak{m}} n_{\mathfrak{m}}$, then there would be some M_i so that $(M_i)_{\mathfrak{m}} = (M_{i+1})_{\mathfrak{m}}$ for all $\mathfrak{m}$. That is, $(M_{i+1}/M_i)_{\mathfrak{m}} = 0$ for all $\mathfrak{m}$, hence $M_{i+1}/M_i = 0$, hence $M_{i+1} = M_i$, a contradiction.
- (c) An example is the $\mathbb{Z}$ -module $\bigoplus_{p \text{ prime}} \mathbb{Z}/p\mathbb{Z}$. Let S be the set of integers not divisible by p , then localizing at S will kill every factor except $\mathbb{Z}/p\mathbb{Z}$, which is finite length as a $\mathbb{Z}[1/S]$ -module. However, the whole module is clearly not finite length.